

Introduction and History of SEM

Agenda

- Introductions
- Overview
- A brief history of SEM
- Review of basic concepts
- Matrix Algebra

Introductions

- Name
- Pronouns (optional)
- University/Department/Area
- Why are you taking this course?
- A boring fact about yourself

Introduction to Structural Equation Modeling

What:

- Multivariate Analysis with Observed Variables
- Multivariate Analysis with Latent Variables
- Advanced applications of the above

How:

- Lecture
- In-class activities
- Homework Assignments
- PRACTICE!

Prerequisites

- Strong understanding of multiple regression analysis
- Experience in R
 - Load in data from csv
 - Transform and work with dataframes
 - Conduct simple statistical analyses (e.g., multiple regression, t-test)
 - Basic syntax (e.g., Boolean logic, matrices and vectors, strings, factors)

Learning Materials

Bollen, K. A. (2026). *Elements of Structural Equation Models (SEMs)*. Cambridge.

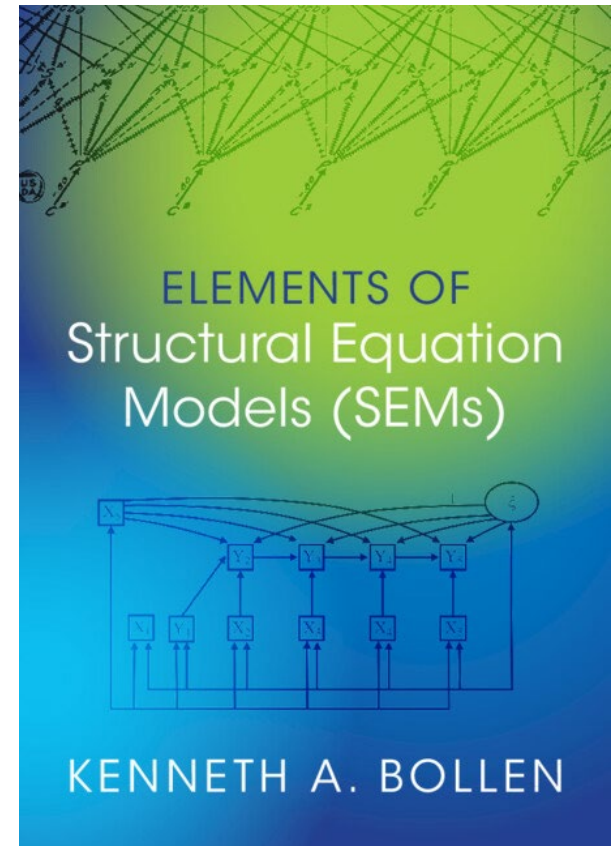
- Don't need for class, but is useful supplement
- Purchase (Cambridge or Amazon)

R

- R is freely downloaded from <https://cran.r-project.org>
- I recommend using Rstudio

Lavaan

- Lavaan is a package in R
- [Documentation](#)



Learning Outcomes

By the end of this class you will be able to...

- Specify, fit, evaluate, and interpret structural equation models which correspond to research hypotheses about the connections among multiple latent and/or observed variables.
- Translate among path diagrams, structural equations (both individual observation and matrix specification), and computer code corresponding to the above types of models.
- Explain connections between other common statistical models (e.g., independent groups t-test, multiple regression, factor analysis) and structural equation models.
- Identify and critique common malpractices in examples of structural equation models.
- Generate written reports which use structural equation models in novel datasets to test hypotheses in your field of study.

List of Topics

- Introduction
- Basic content review (correlation/covariance)
- Matrix Algebra
- Model Notation
- Multiple Regression as an SEM
- Path Analysis
- Observed Variable Identification
- Observed Variable Estimation
- Getting Started with Lavaan
- Observed Variable Model Fit
- Understanding Measurement Error
- Latent Variable Models Specification
- Latent Variable Models Identification
- Latent Variable Models Evaluation
- Combining Models
- Reporting SEMs

First time teaching workshop in 5-day structure, so timing may vary

Questions?

History and Overview of Structural Equation Models

Comparing Methods

Multiple Regression Analysis

- Only one outcome
- All predictors allowed to be freely correlated
- Outcomes and predictors assumed to be measured without error
- Predictors have no implied distribution
- Allows for only observed variables

Structural Equation Modeling

- Multiple outcomes possible
- Covariance among predictors is controllable
- Outcomes and predictors may have measurement error
- (Typically) assume multivariate normality of predictors and outcome
- Allows for observed and latent variables

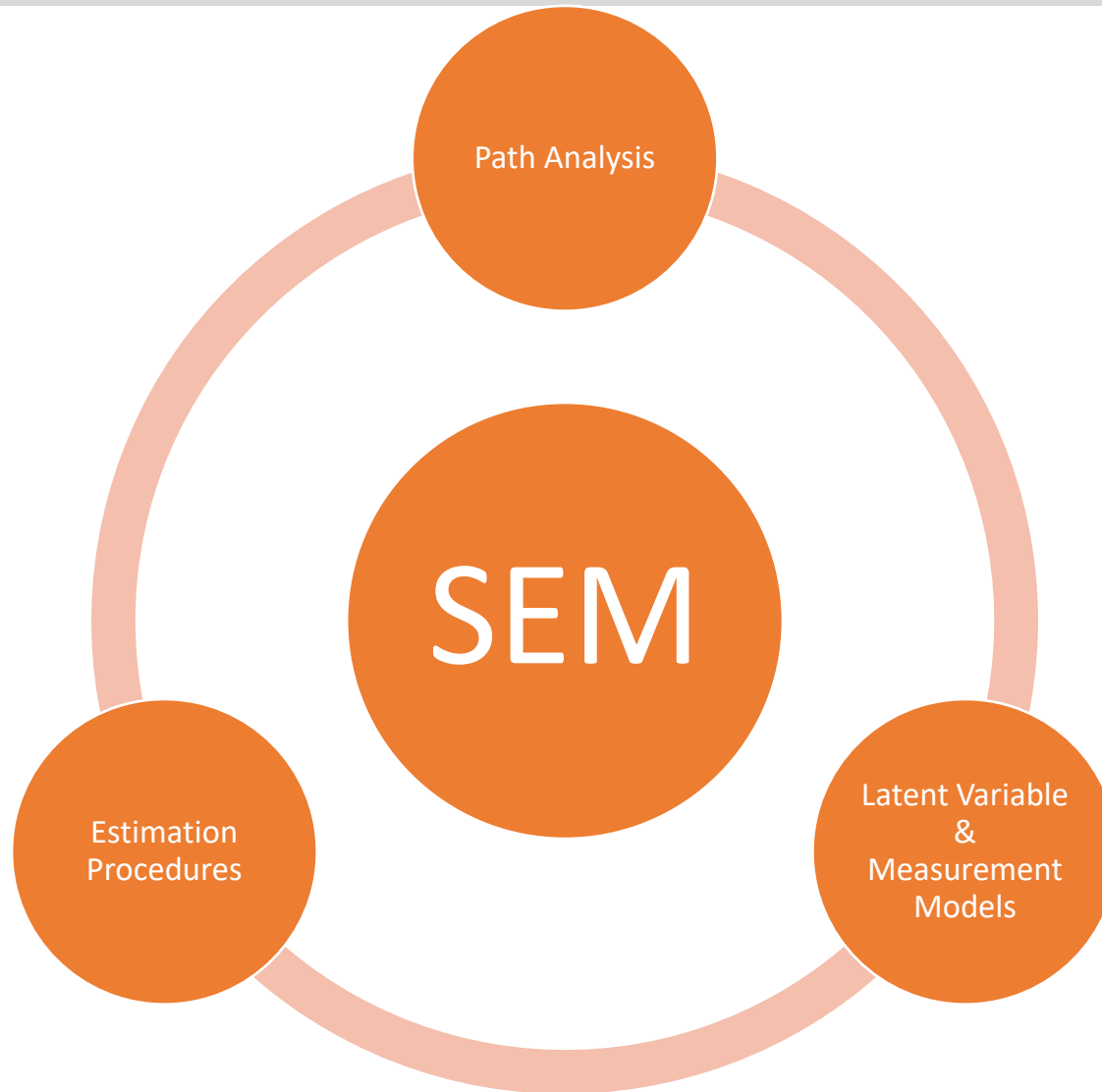
SEM subsumes many other analyses/techniques

- Linear regression
- Confirmatory factor analysis
- Path analysis
- Simultaneous equation systems
- Canonical correlations
- Panel Data Analysis
- ANOVA
- ANCOVA
- Multiple indicator models

So why are there all these other things?

History

Three Major Components to SEM



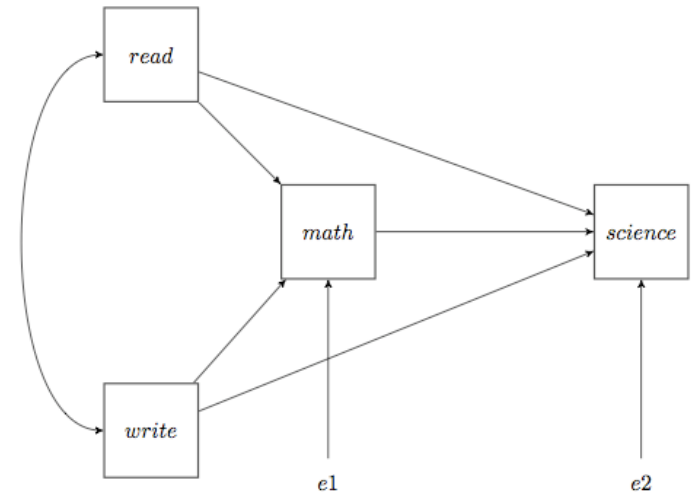
Path Analysis

Sewall Wright outlined a method for visually and mathematically specifying multivariate models (1918-1960)

Defining characteristics:

- **path diagram:** a pictorial representation of a system of simultaneous equations
- Equations linking correlation/covariance to model parameters

Did not use matrix approach (like we will use in this class)



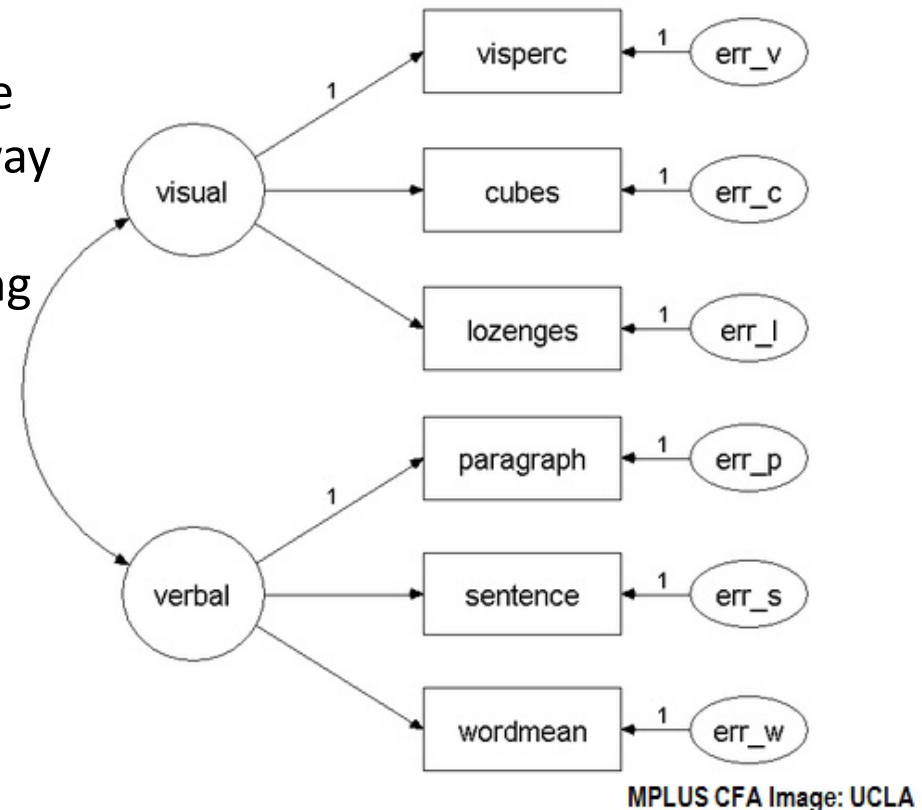
While this method was developed in the early 1900s, it was not widely adopted until the 1960s/70s

- Duncan (1966, 1975) sociology
- Jöreskog (1973) psychology

Latent Variable and Measurement Models

Factor analysis originated with Spearman (1904)

- The idea that there were latent (unobservable) traits, which cause individuals to respond a certain way on observable items
- Much history in intelligence testing (also fraught with eugenics)
- Factor analysis was unconcerned with structural paths between latent variables, just correlations.



Simultaneous Equation Models

Developed in economics (1950s/1960s)

Systems of equations among observed variables which might be measured with error. Did not incorporate latent variables

Primarily developed in a matrix context.

A common example:

- Supply and demand:

Q_t^d : Quantity demanded at time t

Q_t^s : Quantity supplied at time t

P_t : Price at time t

u_t : Measurement errors

$$Q_t^d = \alpha_0 + \alpha_1 P_t + u_{1t} \text{ where } \alpha_1 < 0$$

$$Q_t^s = \beta_0 + \beta_1 P_t + u_{2t} \text{ where } \beta_1 > 0$$

$$Q_t^d = Q_t^s$$

Estimation Procedures

Needed to generate parameter estimates with known properties. Allows you to do things like hypothesis testing

- Maximum likelihood (Jöreskog, 1973)
- Generalized least squares (Jöreskog & Goldberger, 1972; Browne, 1974, 1982, 1984)
- Generalizations to ordinal variables (Muthén, 1984, 1987)

Software development

- LISREL (Jöreskog & Sörbom, 1986a, 1986b, 1988)
- EQS (Bentler, 1985)
- Mplus (Muthen & Muthen, 1998)
- Lavaan (Rosseel, 2012)

Review of Basic Concepts

Building Blocks of SEM

Structural equation models are not fit to individual data.

Instead the model is fit to a covariance matrix or correlation matrix.

In this section we'll review the elements of these matrices

- Covariance
- Variance
- Correlation

Basic Notation: Variables

Two types of variables in this class:

Observed: Z

Four ways of expressing

1. Z a matrix of all Z variables for all cases
2. Z_1 a vector of all cases on the first Z variable
3. Z_{1i} a scalar value of Z_1 for case i
4. $Z_{.i}$ all Z variables for case i (rarely used)

Latent: L

Same ways of expressing

$L, L_1, L_{1i}, L_{.i}$

Basic Notation: Means, Variance, Covariance

Means: μ

Can be used to describe all means of one type of variable (μ_L, μ_Z)

Single mean of a single variable (μ_{Z_2})

Variance & Covariance

As we will discuss, variances and covariances are closely related and will share notation

Σ represents a covariance matrix (variances on diagonal, covariances in off diagonal). Σ_{LL} and Σ_{ZZ} represent covariance matrices for specific variable sets.

σ is used to represent elements of Σ (e.g., $\sigma_{L_1 Z_2}$ is the covariance between L_1 and Z_2)

For shorthand $\sigma_{Z_2}^2$ is the variance of Z_2 (i.e., $\sigma_{Z_2 Z_2}$)

Sometimes we may use $C(\cdot)$ or $V(\cdot)$ for covariance and variance respectively, particularly in derivations

Covariance

Population Covariance

$$\sigma_{Z_1 Z_2} = C(Z_1, Z_2) = \frac{\sum_{i=1}^N (Z_{1i} - \mu_{Z_1})(Z_{2i} - \mu_{Z_2})}{N}$$

Z_{1i} individual i 's score on Z_1

μ_{Z_1} population mean of Z_1

Z_{2i} individual i 's score on Z_2

μ_{Z_2} population mean of Z_2

N sample size

$(Z_{1i} - \mu_{Z_1})$ is the deviation of person i away from the mean on Z_1

$(Z_{2i} - \mu_{Z_2})$ is the deviation of person i away from the mean on Z_2

Positive deviations = higher than the mean

Negative deviations = lower than the mean

Covariance

Population Covariance

$$\sigma_{Z_1 Z_2} = C(Z_1, Z_2) = \frac{\sum_{i=1}^N (Z_{1i} - \mu_{Z_1})(Z_{2i} - \mu_{Z_2})}{N}$$

Z_{1i} individual i 's score on Z_1

μ_{Z_1} population mean of Z_1

Z_{2i} individual i 's score on Z_2

μ_{Z_2} population mean of Z_2

N sample size

We take the product of the deviations to understand the patterns between Z_1 and Z_2

Positive $\times$ *Positive* = *Positive*

High on Z_1 means high on Z_2

Negative $\times$ *Negative* = *Positive*

Low on Z_1 means low on Z_2

Positive $\times$ *Negative* = *Negative*

High on Z_1 means low on Z_2

Negative $\times$ *Positive* = *Negative*

Low on Z_1 means high on Z_2

Covariance

Population Covariance

$$\sigma_{Z_1 Z_2} = C(Z_1, Z_2) = \frac{\sum_{i=1}^N (Z_{1i} - \mu_{Z_1})(Z_{2i} - \mu_{Z_2})}{N}$$

Z_{1i} individual i 's score on Z_1

μ_{Z_1} population mean of Z_1

Z_{2i} individual i 's score on Z_2

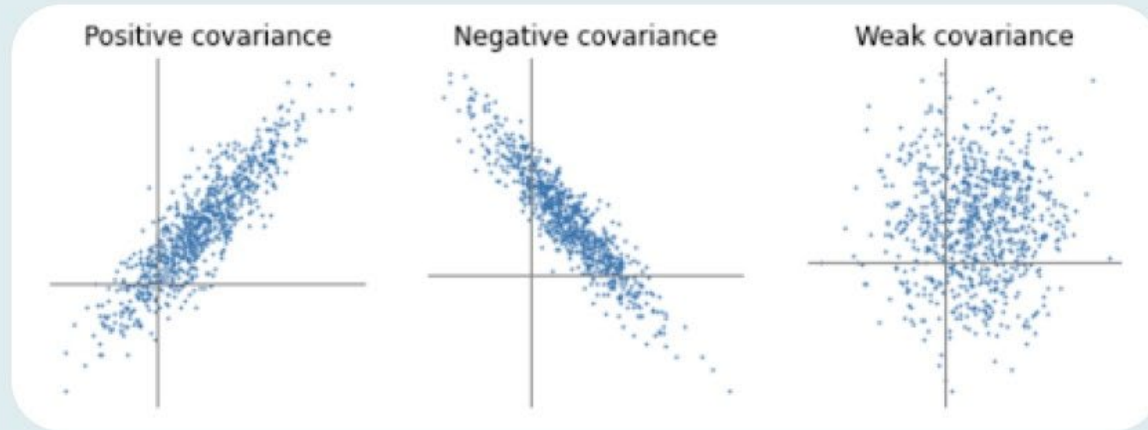
μ_{Z_2} population mean of Z_2

N sample size

We sum up the deviation across each individual to see the general pattern (typically positive or negative)

We divide by N to account for summing across N things, so the estimate is unaffected by sample size

Visualization of Covariance



Whether a covariance is positive, negative, or zero tells us how variables typically move together.

Covariance cannot tell us whether one variable causes another, or whether they covary due to external influence

Variance

Variance is the covariance of a variable with itself

$$\sigma_{Z_1}^2 = V(Z_1) = \frac{\sum_{i=1}^N (Z_{1i} - \mu_{Z_1})(Z_{1i} - \mu_{Z_1})}{N}$$

The two deviations are always the same sign (positive or negative)

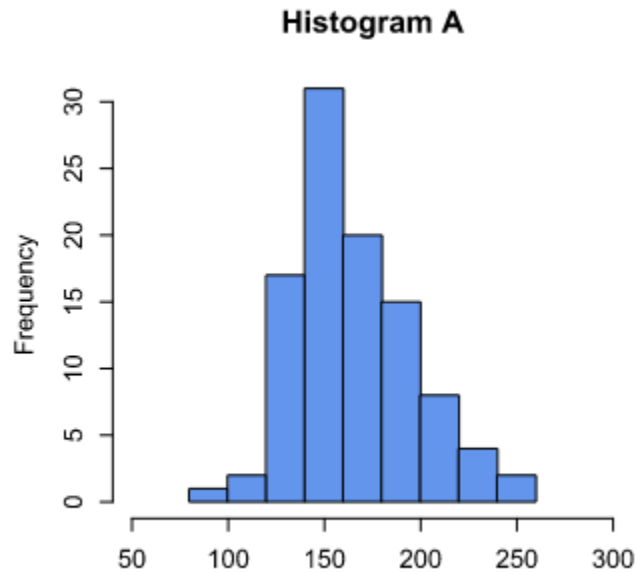
The product of the deviations will always be positive

Positive $\times$ *Positive* = *Positive*

Negative $\times$ *Negative* = *Positive*

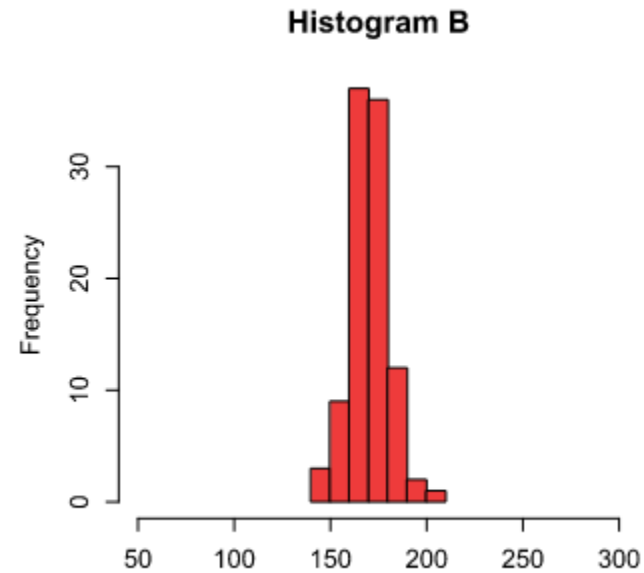
For variance we care more about **magnitude**. Are the deviations from the mean small or large?

Variance Visualization



Higher Variance

Points deviate further from the mean on average



Lower Variance

Points deviate only a little from the mean

Estimating Variance and Covariance

Typically we do not know the parameters μ_{Z_1} and μ_{Z_2}

Instead we must estimate these parameters with the sample means

$$\widehat{\mu}_{Z_1} = \overline{Z_1} \text{ and } \widehat{\mu}_{Z_2} = \overline{Z_2}$$

Therefore we estimate covariance as

$$\hat{\sigma}_{Z_1 Z_2} = \hat{C}(Z_1, Z_2) = \frac{\sum_{i=1}^N (Z_{1i} - \overline{Z_1})(Z_{2i} - \overline{Z_2})}{N - 1}$$

and variance as

$$\hat{\sigma}_{Z_1}^2 = \hat{V}(Z_1) = \frac{\sum_{i=1}^N (Z_{1i} - \overline{Z_1})(Z_{1i} - \overline{Z_1})}{N - 1}$$

Why $N - 1$?

When we do not know the means of the variables, and we must estimate them, we use 1 degree of freedom in that estimation.

This is an **unbiased** estimate of population variance.

Correlation

Correlation can be thought about two ways:

- 1) Correlation is the covariance of standardized variables

Assume

$$Z_{1i}^s = (Z_{1i} - \bar{Z}_1) / \hat{\sigma}_{Z_1} \text{ and } Z_{2i}^s = (Z_{2i} - \bar{Z}_2) / \hat{\sigma}_{Z_2}$$

Then the correlation between Z_1 and Z_2 is the covariance between Z_1^s and Z_2^s

$$\hat{\rho}_{Z_1 Z_2} = r_{Z_1 Z_2} = \hat{C}(Z_1^s, Z_2^s) = \frac{\sum_{i=1}^N (Z_{1i}^s - \bar{Z}_{1i}^s)(Z_{2i}^s - \bar{Z}_{2i}^s)}{N - 1}$$

But because Z_{1i}^s and Z_{2i}^s are standardized variables, we know their means are zero

$$\hat{\rho}_{Z_1 Z_2} = r_{Z_1 Z_2} = \hat{C}(Z_1^s, Z_2^s) = \frac{\sum_{i=1}^N (Z_{1i}^s)(Z_{2i}^s)}{N - 1}$$

Correlation

Correlation can be thought about two ways:

- 2) Correlation can be calculated from the variance and covariance of each variable

$$\hat{\rho}_{Z_1 Z_2} = r_{Z_1 Z_2} = \frac{\hat{\sigma}_{Z_1 Z_2}}{\hat{\sigma}_{Z_1} \hat{\sigma}_{Z_2}}$$

Correlation is the covariance of Z_1 and Z_2 , scaled by the standard deviation of Z_1 and Z_2

How do we know these are the same?

$$\frac{\sum_{i=1}^N (Z_{1i}^S)(Z_{2i}^S)}{N-1}$$

$$\frac{\sum_{i=1}^N \left(\frac{Z_{1i} - \bar{Z}_1}{\hat{\sigma}_{Z_1}} \right) \left(\frac{Z_{2i} - \bar{Z}_2}{\hat{\sigma}_{Z_2}} \right)}{N-1}$$

$$\frac{\sum_{i=1}^N (Z_{1i} - \bar{Z}_1)(Z_{2i} - \bar{Z}_2)}{(N-1)\hat{\sigma}_{Z_1}\hat{\sigma}_{Z_2}}$$

$$\frac{\hat{\sigma}_{Z_1 Z_2}}{\hat{\sigma}_{Z_1} \hat{\sigma}_{Z_2}}$$

we know $Z_{1i}^S = (Z_{1i} - \bar{Z}_1)/\hat{\sigma}_{Z_1}$ and $Z_{2i}^S = (Z_{2i} - \bar{Z}_2)/\hat{\sigma}_{Z_2}$

Let's move the estimated SDs ($\hat{\sigma}_{Z_1}$ and $\hat{\sigma}_{Z_2}$) to the denominator

But we know that

$$\hat{\sigma}_{Z_1 Z_2} = \frac{\sum_{i=1}^N (Z_{1i} - \bar{Z}_1)(Z_{2i} - \bar{Z}_2)}{N-1}$$

This is our second equation for correlation

Covariance, Variance, Correlation Summary

What?	Population	Sample (Greek)	Sample (lower case)
Covariance (Z_1, Z_2)	$\sigma_{Z_1 Z_2} = C(Z_1, Z_2)$ $= \frac{\sum_{i=1}^N (Z_{1i} - \mu_{Z_1})(Z_{2i} - \mu_{Z_2})}{N}$	$\hat{\sigma}_{Z_1 Z_2} = \hat{C}(Z_1, Z_2)$ $= \frac{\sum_{i=1}^N (Z_{1i} - \bar{Z}_1)(Z_{2i} - \bar{Z}_2)}{N - 1}$	$s_{Z_1 Z_2} = \frac{\sum_{i=1}^N (Z_{1i} - \bar{Z}_1)(Z_{2i} - \bar{Z}_2)}{N - 1}$
Variance (Z_1)	$\sigma_{Z_1}^2 = V(Z_1)$ $= \frac{\sum_{i=1}^N (Z_{1i} - \mu_{Z_1})(Z_{1i} - \mu_{Z_1})}{N}$	$\hat{\sigma}_{Z_1}^2 = \hat{V}(Z_1)$ $= \frac{\sum_{i=1}^N (Z_{1i} - \bar{Z}_1)(Z_{1i} - \bar{Z}_1)}{N - 1}$	$s_{Z_1}^2 = \frac{\sum_{i=1}^N (Z_{1i} - \bar{Z}_1)(Z_{1i} - \bar{Z}_1)}{N - 1}$
Correlation (Z_1, Z_2)	$\rho_{Z_1 Z_2} = \frac{\sigma_{Z_1 Z_2}}{\sigma_{Z_1} \sigma_{Z_2}}$	$\hat{\rho}_{Z_1 Z_2} = \frac{\widehat{\sigma_{Z_1 Z_2}}}{\hat{\sigma}_{Z_1} \hat{\sigma}_{Z_2}}$	$r_{Z_1 Z_2} = \frac{s_{Z_1 Z_2}}{s_{Z_1} s_{Z_2}}$

Some notes on notation:

- Different books, articles, etc. will use different notation so I want to expose you to some of the most common
- There does not seem to be a common lower case notation for covariance (let's generalize to use $s_{Z_1 Z_2}$)
- C and V notations can often be ambiguous as to whether they are sample or population
- Hats universally mean estimate (i.e., sample)

Covariance Matrix

The major input of an SEM is a covariance matrix.

Let's say we have 3 variables (Z_1, Z_2, Z_3). The population covariance matrix is...

$$\Sigma = \begin{bmatrix} \sigma_{Z_1}^2 & \sigma_{Z_1 Z_2} & \sigma_{Z_1 Z_3} \\ \sigma_{Z_1 Z_2} & \sigma_{Z_2}^2 & \sigma_{Z_2 Z_3} \\ \sigma_{Z_1 Z_3} & \sigma_{Z_2 Z_3} & \sigma_{Z_3}^2 \end{bmatrix}$$

This matrix is a **3x3** matrix (3 rows, 3 columns) [NOTE: always rows and columns in that order]

The matrix is considered **square** because it has an equal number of rows and columns

Variances are along the diagonal

Covariances are on the off diagonals

The matrix is described as **symmetric** because the i,j^{th} element is the same as the j,i^{th} element [Remember rows, column]

Covariance Matrix

The population covariance matrix is...

$$\Sigma = \begin{bmatrix} \sigma_{Z_1}^2 & \sigma_{Z_1 Z_2} & \sigma_{Z_1 Z_3} \\ \sigma_{Z_1 Z_2} & \sigma_{Z_2}^2 & \sigma_{Z_2 Z_3} \\ \sigma_{Z_1 Z_3} & \sigma_{Z_2 Z_3} & \sigma_{Z_3}^2 \end{bmatrix}$$

The sample covariance matrix is...

$$\mathbf{S} = \begin{bmatrix} s_{Z_1}^2 & s_{Z_1 Z_2} & s_{Z_1 Z_3} \\ s_{Z_1 Z_2} & s_{Z_2}^2 & s_{Z_2 Z_3} \\ s_{Z_1 Z_3} & s_{Z_2 Z_3} & s_{Z_3}^2 \end{bmatrix}$$

For convenience, with a symmetric matrix we may drop the upper off diagonal....

$$\mathbf{S} = \begin{bmatrix} s_x^2 & & \\ s_{xy} & s_y^2 & \\ s_{xz} & s_{yz} & s_z^2 \end{bmatrix}$$

Correlation Matrix

Early forms of SEM would take as input a correlation matrix

With 3 variables (Z_1, Z_2, Z_3). The population correlation matrix is...

$$\mathbf{P} = \begin{bmatrix} 1 & \rho_{Z_1 Z_2} & \rho_{Z_1 Z_3} \\ \rho_{Z_1 Z_2} & 1 & \rho_{Z_2 Z_3} \\ \rho_{Z_1 Z_3} & \rho_{Z_2 Z_3} & 1 \end{bmatrix}$$

This matrix is a **3x3** matrix (3 rows, 3 columns) [NOTE: always rows and columns in that order]

The matrix is considered **square** because it has an equal number of rows and columns

1s are along the diagonal

correlations are on the off diagonals

The matrix is described as **symmetric** because the i,j^{th} element is the same as the j,i^{th} element [Remember rows, column]

Note: **P** (upper case rho) notation is not in the book

Summary

- SEM is a general modeling tool that combines path analysis, measurement models, and advanced estimation procedures
 - Many models you have encountered previously are special cases of SEM
- Correlation, Covariances, and Variances describe the relationship between variables and the variability within variables used in an SEM analysis
- Matrices storing these values are the “data” upon which SEM is based
- Appropriate notation is necessary to distinguish population and sample values of important statistics (correlation, covariance, variance)

References

- Wright, S. (1918). On the nature of size factors. *Genetics*, 3, 367 – 374.
- Wright, S. (1921). Correlation and causation. *Journal of Agricultural Research*, 20, 557 – 585.
- Wright, S. (1935). The method of path coefficients. *Annals of Mathematical Statistics*, 5, 161 – 215.
- Wright, S. (1954). The interpretation of multivariate systems. In O. K. Kempthorne, T. A. Bancroft, J. W. Gowen, and J. L. Lush, eds., *Statistics and Mathematics in Biology*. Ames: Iowa State University Press, pp. 11 – 33.
- Wright, S. (1960). Path coefficients and path regressions: Alternative or complementary concepts? *Biometrics*, 16, 189 – 202.
- Duncan, O. D. (1966). Path analysis: Sociological examples. *American Journal of Sociology*, 72, 1 – 16.
- Duncan, O. D. (1975). *Introduction to Structural Equation Models*. New York: Academic Press.
- Jöreskog, K. G. (1973). A general method for estimating a linear structural equation system. In A. S. Goldberger and O. D. Duncan, eds., *Structural Equation Models in the Social Sciences*. New York: Academic Press, pp. 85 – 112.
- Spearman, C. (1904). General intelligence, objectively determined and measured. *American Journal of Psychology*, 15, 201 – 293.
- Jöreskog, K. G., & Goldberger, A. S. (1972). Factor analysis by generalized least squares. *Psychometrika*, 37, 243 – 260.
- Browne, M. W. (1974). Generalized least-squares estimators in the analysis of covariance structures. *South African Statistical Journal*, 8, 1 – 24.
- Browne, M. W. (1982). Covariance structures. In D. M. Hawkins, ed., *Topics in Multivariate Analysis*. Cambridge University Press, pp 72 – 141.
- Browne, M. W. (1984). Asymptotic distribution free methods in analysis of covariance structures. *British Journal of Mathematical and Statistical Psychology*, 37, 62 – 83.
- Bentler, P. M. (1983). Simultaneous equation systems as moment structure models. *Journal of Econometrics*, 22, 13 – 42.
- Muthén, B. (1984). A general structural equation model with dichotomous, ordered categorical and continuous latent variable indicators. *Psychometrika*, 49, 115 – 132.
- Muthén, B. (1987). *LISCOMP: Analysis of Linear Structural Equations with a Comprehensive Measurement Model*. Mooresville, IN: Scientific Software, Inc.
- Jöreskog, K. G. & Sörbom, D. (1986a). *PRELIS: A preprocessor for LISREL*. Mooresville, IN: Scientific Software Inc.
- Jöreskog, K. G. & Sörbom, D. (1988). *LISREL 7: A Guide to the Program and Applications*. Chicago: SPSS, Inc.
- Bentler, P. M. (1985). Theory and implementation of EQS: A structural equations program. Los Angeles: BMDP Statistical Software.
- Muthén, L.K. and Muthén, B.O. (1998-2017). *Mplus User's Guide*. Eighth Edition. Los Angeles, CA: Muthén & Muthén
- Rosseel Y (2012). "lavaan: An R Package for Structural Equation Modeling." *Journal of Statistical Software*, 48(2), 1–36. doi:10.18637/jss.v048.i02.